

Module 13

Section 8.4

Vectors

Math 130: Advanced Technical Mathematics | Unit 3

Why Vectors?

Many real-world quantities need both size AND direction.

- Velocity — wind at 15 mph to the north-east
- Force — 50 N pulling down at 30°
- Displacement — walking 3 blocks east, then 2 north

Scalars (speed, mass, temperature) need only size.

Module 13 — Skills Overview

- 1 Writing a vector in component form from initial/terminal points
- 2 Vector addition and scalar multiplication: component form
- 3 Finding magnitude and direction of a vector from its graph
- 4 Finding the direction angle of a vector in $a\mathbf{i} + b\mathbf{j}$ form
- 5 (Supplementary) Finding the components of a vector given magnitude and angle

Vectors and Scalars

Scalar

Magnitude only (a single number)

Examples: speed, temperature, mass

Vector

Magnitude AND direction

Examples: velocity, force, displacement

Notation

Component form: $\mathbf{v} = \langle a, b \rangle = a \mathbf{i} + b \mathbf{j}$

Unit vectors: $\mathbf{i} = \langle 1, 0 \rangle$, $\mathbf{j} = \langle 0, 1 \rangle$

Zero vector: $\mathbf{0} = \langle 0, 0 \rangle$ (no magnitude, no direction)

Magnitude: $|\mathbf{v}| = \sqrt{a^2 + b^2}$ Direction angle: θ (CCW from +x)

Writing a Vector in Component Form

Formula: $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$ (terminal minus initial)

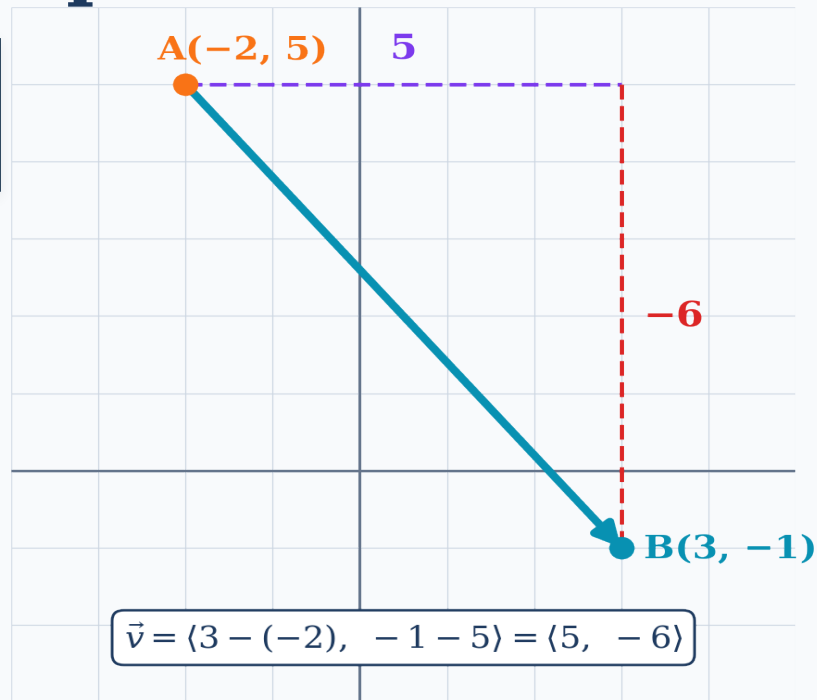
Example: A(-2, 5) to B(3, -1)

$$\mathbf{v} = \langle 3 - (-2), -1 - 5 \rangle$$

$$\mathbf{v} = \langle 5, -6 \rangle$$

Meaning: 5 units right, 6 units down

Component form is position-independent — only the displacement matters.



Vector Addition & Scalar Multiplication

Addition $\langle a, b \rangle + \langle c, d \rangle = \langle a+c, b+d \rangle$

Subtraction $\langle a, b \rangle - \langle c, d \rangle = \langle a-c, b-d \rangle$

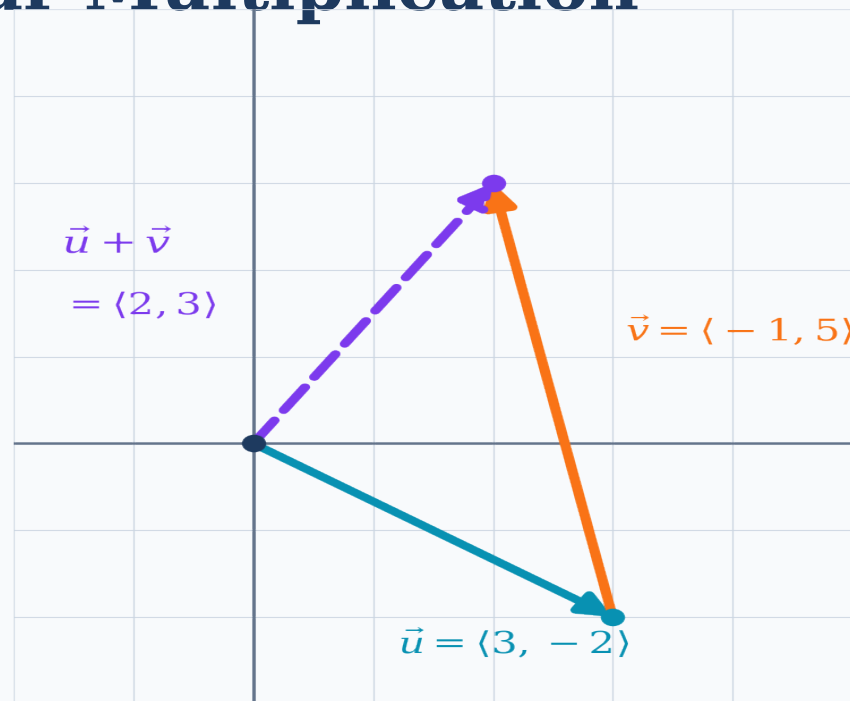
Scalar $\times$ $k\langle a, b \rangle = \langle ka, kb \rangle$

$\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$

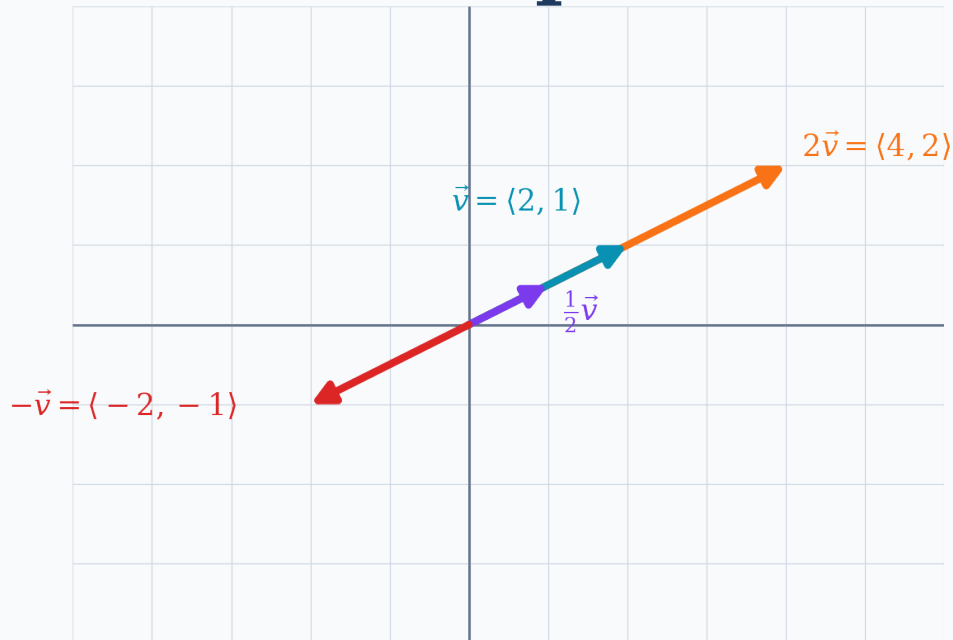
$\mathbf{u} + \mathbf{v} = \langle 3+(-1), -2+5 \rangle = \langle 2, 3 \rangle$

$3\mathbf{u} - 2\mathbf{v}$:

$3\langle 3, -2 \rangle - 2\langle -1, 5 \rangle = \langle 9, -6 \rangle - \langle -2, 10 \rangle$
 $= \langle 9-(-2), -6-10 \rangle = \langle 11, -16 \rangle$



Scalar Multiplication — Visual



Rules

$|k| > 1 \rightarrow$ stretches the vector

$0 < |k| < 1 \rightarrow$ shrinks the vector

$k < 0 \rightarrow$ reverses direction

$$k\langle a, b \rangle = \langle ka, kb \rangle$$

Magnitude: $|k\vec{v}| = |k| \times |\vec{v}|$

Magnitude and Direction

Magnitude: $|\mathbf{v}| = \sqrt{a^2 + b^2}$

Direction: $\theta = \tan^{-1}(b / a)$
(adjust for quadrant)

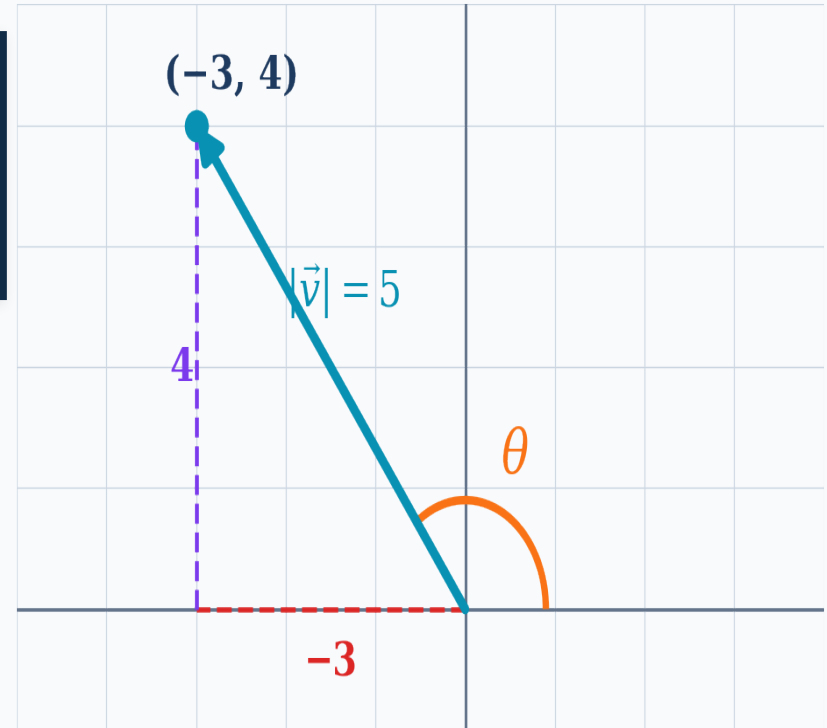
Example: $\mathbf{v} = \langle -3, 4 \rangle$

$$|\mathbf{v}| = \sqrt{9 + 16} = \sqrt{25} = 5$$

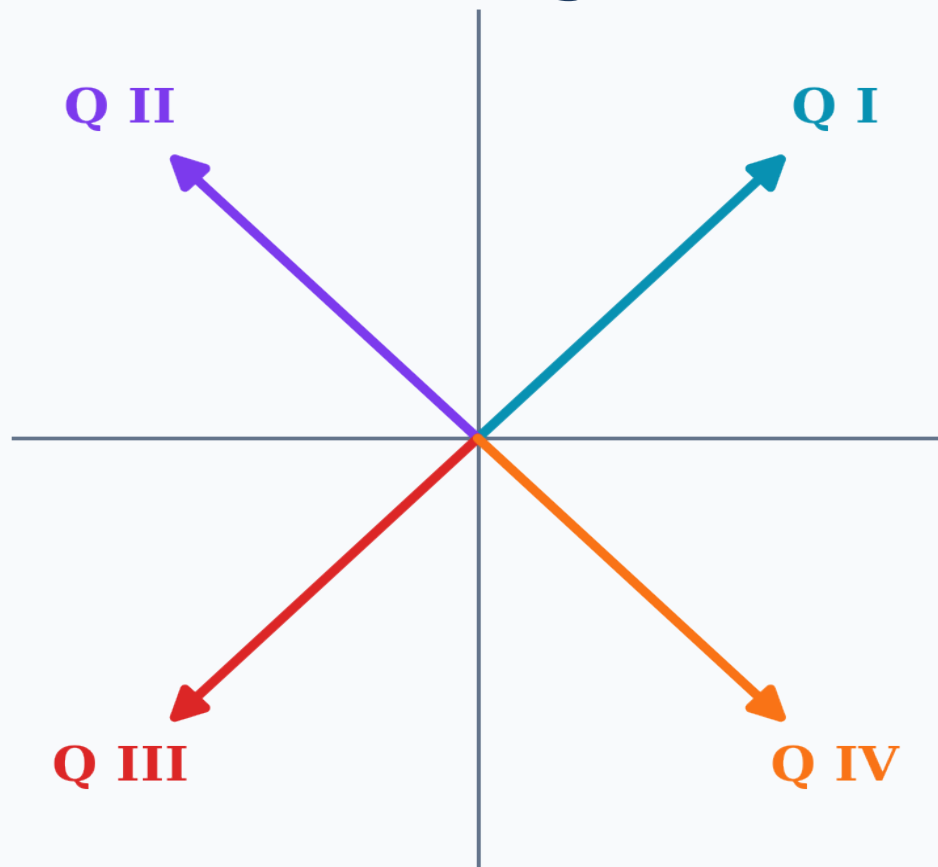
$$\tan^{-1}(4/-3) \approx -53.1^\circ$$

Q II ($a < 0, b > 0$): add 180°

$$\theta \approx -53.1^\circ + 180^\circ = 126.9^\circ$$



Direction Angle — Quadrant Adjustment



Examples

$\langle 3, 4 \rangle$ Q I

$\theta \approx 53.1^\circ$

$\langle -5, 12 \rangle$ Q II

$\theta \approx 112.6^\circ$

$\langle -3, -7 \rangle$ Q III

$\theta \approx 246.8^\circ$

$\langle 4, -2 \rangle$ Q IV

$\theta \approx 333.4^\circ$

Direction Angle of $ai + bj$

Find the direction angle of $v = -5i + 12j$.

Convert to components:

$$v = -5i + 12j \rightarrow \langle -5, 12 \rangle \quad (a = -5, b = 12)$$

Find magnitude:

$$|v| = \sqrt{(25 + 144)} = \sqrt{169} = 13$$

Find direction:

$$\tan^{-1}(12/-5) \approx -67.4^\circ$$

$a < 0, b > 0 \rightarrow$ Q II: add 180°

$$\theta \approx -67.4^\circ + 180^\circ = 112.6^\circ$$

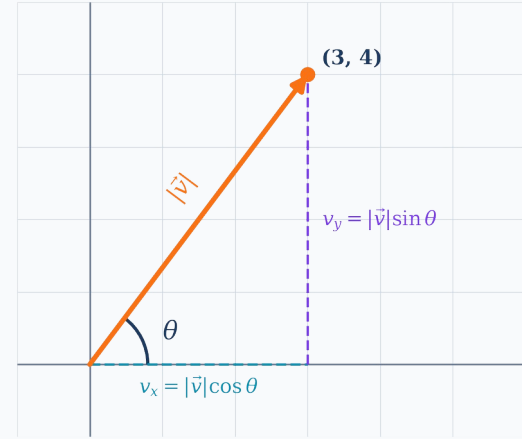
Components from $|\mathbf{v}|$ and θ

Given $|\mathbf{v}|$ and θ

- $v_x = |\mathbf{v}| \cos \theta$

$$v_y = |\mathbf{v}| \sin \theta$$

$$\mathbf{v} = \langle |\mathbf{v}| \cos \theta, |\mathbf{v}| \sin \theta \rangle$$



Example: $|\mathbf{v}| = 5$, $\theta = 53.1^\circ$

$$v_x = 5 \cos 53.1^\circ \approx 3$$

$$v_y = 5 \sin 53.1^\circ \approx 4$$

$$\mathbf{v} = \langle 3, 4 \rangle$$

You Try

Vector v goes from $P(-4, -2)$ to $Q(2, 1)$.

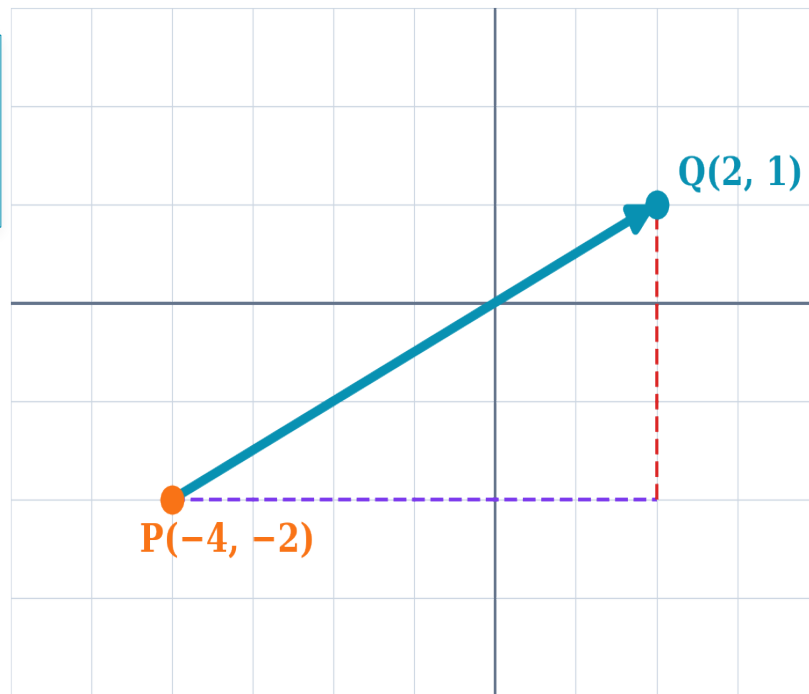
Find:

- (a) component form
- (b) magnitude
- (c) direction angle

(a) $v = \langle 2 - (-4), 1 - (-2) \rangle = \langle 6, 3 \rangle$

(b) $|v| = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5} \approx 6.71$

(c) $\theta = \tan^{-1}(3 / 6) \approx 26.57^\circ$ (Q I, no adjustment)



Test Prep — Vector Operations

Test 3, Problem 9: Vector addition, scalar multiplication, magnitude, and direction angle calculations.

Part A: Component form

Given: $P(3, -1)$ to $Q(-2, 4)$

$$\mathbf{v} = \langle -2-3, 4-(-1) \rangle$$

$$\mathbf{v} = \langle -5, 5 \rangle$$

$$|\mathbf{v}| = \sqrt{(25+25)} = \sqrt{50} = 5\sqrt{2}$$

$$\theta = \tan^{-1}(5/-5) + 180^\circ = -45^\circ + 180^\circ$$

$$\theta = 135^\circ$$

Part B: Operations

Given: $\mathbf{u} = \langle 2, -3 \rangle$, $\mathbf{v} = \langle -1, 4 \rangle$

$$\mathbf{u} + \mathbf{v} = \langle 1, 1 \rangle$$

$$2\mathbf{u} - 3\mathbf{v} =$$

$$\langle 4, -6 \rangle - \langle -3, 12 \rangle = \langle 7, -18 \rangle$$

$$|2\mathbf{u} - 3\mathbf{v}| = \sqrt{(49+324)} = \sqrt{373} \\ \approx 19.3$$

Module 13 Summary

5 ALEKS Topics | Section 8.4

Components

$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$ from points, or $\mathbf{v} = \langle |\mathbf{v}|\cos \theta, |\mathbf{v}|\sin \theta \rangle$ from magnitude/direction

Operations

Add/subtract component-wise, scalar multiplication multiplies each component

Magnitude & Direction

$|\mathbf{v}| = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$ adjusted for quadrant (Q II/III: $+180^\circ$, Q IV: $+360^\circ$)

Key for Test 3

Problem 9: component form, vector operations ($2\mathbf{u} - 3\mathbf{v}$), magnitude, and direction angle with quadrant adjustment

Next: Unit 3 Review → Test 3